

AI-Assisted Clinical Diagnostics Platform

A cloud-native digital-health platform that wraps a deep-learning image model in a safe, compliant clinical workflow — quality-gated inference, deterministic risk banding, mandatory provider attestation, and a research harness that proves the model against clinical ground truth.

Deep-learning inference

Model-as-a-service

Risk banding

Image-quality gate

Human-in-the-loop

Sensitivity / specificity

FastAPI

Next.js

Express + Puppeteer

Clerk SSO

DocumentDB

AWS App Runner

Terraform IaC

HIPAA-conscious

Turn a research model into a safe, compliant, provable clinical product

A digital-health company set out to deliver **AI-assisted diagnostic screening for a common acute condition** through at-home capture and clinician review. Turning a deep-learning classifier into a regulated product raised several hard problems at once.

A model is not a product

Raw model outputs had to become calibrated, explainable, audience-appropriate guidance — never a bare probability handed to a worried patient.

Human oversight is non-negotiable

For a regulated use case, AI can assist but a licensed provider must review and attest to every result; unusable inputs must fail safe as "inconclusive", not guess.

It handles PHI end-to-end

Clinical images, symptoms, and identities are protected health information — consent explicit and revocable, storage encrypted, access role-scoped, actions audited.

It has to prove itself — and keep evolving

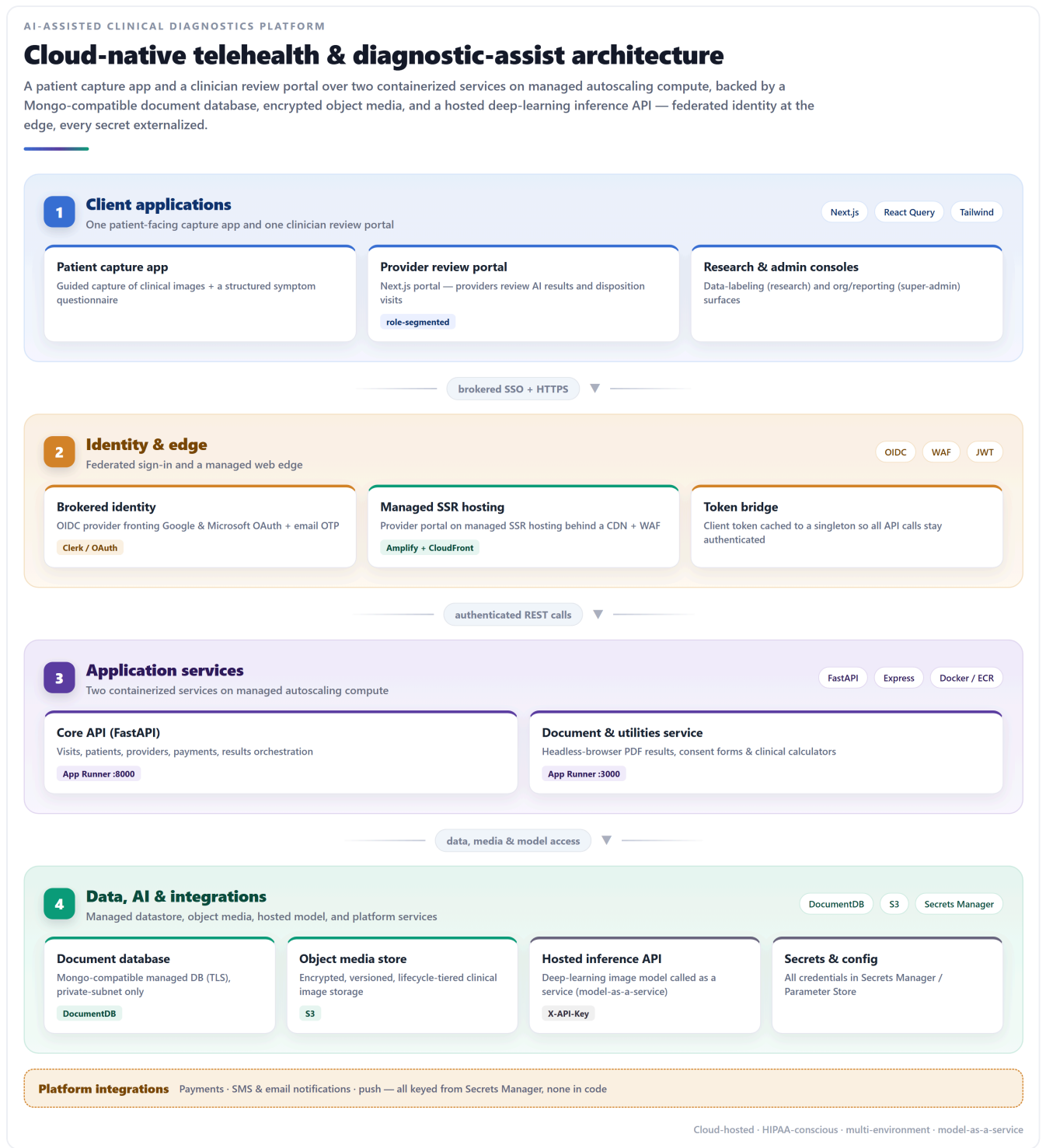
A clinical AI must demonstrate sensitivity, specificity and predictive value against real ground truth, across a model that would be re-versioned repeatedly, on repeatable cloud infrastructure.

AI assists, a clinician decides — and the model is measured, not assumed

A four-service platform on AWS, engineered so the model is one safe component inside a compliant, observable, continuously-improvable clinical workflow.

- ✓ **Safe-by-construction pipeline.** Guided capture + structured intake → an OpenCV **image-quality gate** (fail-safe inconclusive) → a **hosted deep-learning model** → a deterministic **three-tier risk band** with separate patient/provider copy and a standard clinical score → mandatory **provider attestation**.
- ✓ **Model-as-a-service.** Inference is a hosted HTTP model behind an API key, decoupled from the app — re-versioned and scaled independently (migrated from earlier in-process PyTorch/YOLO).
- ✓ **Evidence, not assumptions.** A research harness captures multi-site labeled data; a deterministic clinical rules engine builds a multi-class differential ground truth; the model is scored with confusion-matrix sensitivity / specificity / PPV / NPV across versions.
- ✓ **One portal, three roles, many brands.** A role-segmented Next.js portal (provider / super-admin / research) behind one federated SSO, with white-label theming by organization.
- ✓ **PHI-aware.** Explicit revocable consent with audit trail, role-scoped auth, encrypted private datastores, structured audit logging.
- ✓ **Modular IaC + image-based CI/CD.** One Terraform module per concern across isolated, state-locked dev/staging/prod; CI rolls container images while Terraform owns runtime config.

Cloud-native platform architecture



The AI diagnostic pipeline

AI DIAGNOSTIC PIPELINE

From captured image to clinician-attested result

Guided capture and a structured questionnaire feed an automated quality gate, then a hosted deep-learning model; raw scores are converted by a deterministic three-tier risk-banding rule into audience-aware guidance, and a licensed provider always reviews and attests before a result is issued.

1 Capture & intake

Structured inputs gathered at the point of care

Clinical image capture

Multi-step guided image capture of the assessment site

Symptom questionnaire

Structured symptom & history intake (SOAP-style note)

Consent & PHI release

Per-member consent recorded with authorize/revoke audit trail

PHI consent

automated quality gate

2 Image-quality gating

Reject unusable inputs before they reach the model

Blur / quality check

On-device-style variance check flags low-quality frames

OpenCV

Inconclusive short-circuit

Low-quality input returns an explicit inconclusive state, not a guess

model-as-a-service call

3 Hosted deep-learning inference

A remote image model returns classification + graded findings

hosted model

HTTPS + API key

Classification score

Softmax probability the condition is present

Graded findings

Per-region severity grading of clinical features

Preprocessing flags

Segmentation / reflection-removal quality signals

local decision rule

4 Risk banding & dual presentation

Deterministic banding turns raw scores into safe, audience-aware guidance

Three-tier risk band

Low / Moderate / High from calibrated score thresholds

low · mod · high

Dual audience copy

Distinct patient-facing vs. provider-facing risk presentation

Standard clinical score

A published symptom-based clinical risk score computed alongside

clinician in the loop

5 Provider review & output

A licensed clinician always dispositions the result

AI-in-the-loop review

Provider inspects images, findings, scores & explainability panels

Attestation

Provider attests review & treatment plan before completion

human sign-off

Result PDF

Server-rendered patient/provider results document generated

Quality-gated · model-as-a-service · deterministic banding · human-in-the-loop

Capture → image-quality gate (fail-safe) → hosted model → deterministic risk banding + dual presentation → provider review & attestation → result.

Model evaluation & research harness

MODEL EVALUATION & RESEARCH HARNESS

Proving diagnostic quality against clinical ground truth

A research pipeline captures multi-site labeled data, a deterministic clinical rules engine compiles lab panels and symptoms into a multi-class differential ground truth, and the model is scored with sensitivity, specificity, and predictive-value metrics — versioned and tracked release over release.

A built-in research & evaluation harness measures the model against clinical ground truth

Separate from the live path — how model quality is proven and tracked release over release

1 Labeled data capture

Research assistants collect and quality-tag study data

Study visit capture

Images + structured symptoms captured across multiple sites

Media quality tagging

Each capture graded for usability & completeness

Lab result entry

Confirmatory lab panels recorded per visit

ground-truth source

deterministic labeling

2 Clinical rules engine

Labs + symptoms compiled into a differential ground-truth label

rules engine

differential dx

Rule-based classifier

Hand-built predicates over lab panels & symptom sets

~20 condition codes

Multi-class differential (bacterial / viral / other) ground truth

multi-class

Batch labeling

CSV/batch ground-truth generation for whole datasets

model vs. ground truth

3 Evaluation & versioning

Quantify performance and gate model versions

Confusion matrix

Model predictions scored against ground-truth labels

Sensitivity / specificity

PPV / NPV and recall-threshold metrics computed

clinical metrics

Model versioning

Versioned model + explainability content shipped per release

Multi-site · rules-based ground truth · sensitivity / specificity · versioned

Multi-site labeled capture, a deterministic clinical rules engine for multi-class ground truth, and sensitivity / specificity / PPV / NPV tracked across model versions.

Cloud delivery & platform engineering

CLOUD DELIVERY & PLATFORM ENGINEERING

Modular Terraform IaC with image-based CI/CD

A modular Terraform codebase provisions the whole estate — network, registry, autoscaling compute, managed database, object storage, secrets and least-privilege IAM — across isolated, state-locked dev / staging / production environments, with GitHub Actions rolling container images while Terraform stays the source of truth for config.

1 Modular infrastructure as code

One Terraform module per concern, composed per environment

Terraform

modules

least-privilege IAM

Network & endpoints

VPC interface endpoints, security groups, private subnets

VPC

Registry & compute

ECR repositories + two autoscaling App Runner services

Data & secrets

DocumentDB, S3, Parameter Store, Secrets Manager, IAM

Access proxy

SSM-only bastion for DB access — no SSH ingress

Session Manager

remote state, per environment ▼

2 Multi-environment estate

Isolated dev / staging / production, state-locked

Three environments

dev / staging / prod, each its own state, profile & sizing

dev · stg · prod

Locked remote state

Object-store backend + lock table, encrypted

Prod hardening

Higher autoscaling ceilings + CloudWatch alarms & SNS alerts

image-based continuous delivery ▼

3 CI/CD delivery

Build once, roll the image, keep config in Terraform

ECR

App Runner

Amplify

Build & push

GitHub Actions builds Docker images, pushes to ECR (sha + latest)

GitHub Actions

Roll the service

Pipeline updates the App Runner service, preserving env config

Portal via managed SSR

Provider portal builds & deploys through managed SSR hosting

Modular IaC · multi-environment · image-based CD · config-as-code

Modular Terraform provisions the estate across isolated, state-locked dev / staging / prod, with GitHub Actions rolling container images and Terraform owning config.

A model made safe, compliant, provable — and operable at scale

Human-in-the-loop

Every AI result reviewed & attested by a licensed provider before issue

Fail-safe AI

Image-quality gate + explicit inconclusive state + deterministic risk banding

Sens / spec / PPV / NPV

Model measured against rules-engine clinical ground truth, per version

4 services, 3 envs

FastAPI + Express + Next.js on AWS via modular Terraform, dev/stg/prod

The model became a product. Quality gating, calibrated risk banding, dual patient/provider presentation, a familiar standard clinical score, and a mandatory provider attestation turned a raw deep-learning probability into guidance clinicians trust and patients understand — with model-as-a-service decoupling so the model evolves without touching the app.

And it can prove itself. A dedicated research and evaluation harness — multi-site labeled capture, a deterministic clinical rules engine for multi-class differential ground truth, and confusion-matrix sensitivity/specificity/PPV/NPV tracked across versions — gives the platform the evidence base a regulated clinical AI product requires, all on repeatable, governed AWS infrastructure-as-code.